Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using content-based filtering approach	
Experiment No: 11	**Date:**	**Enrolment No:92301733021**

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab

Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.
In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:
In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.
Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:
The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.
Advantages and Disadvantages:

- **Advantages:**
    - No need for data on other users when applying to similar users.
    - Able to recommend to users with unique tastes.
    - Able to recommend new & popular items

- Explanations for recommended items.
- **Disadvantages**:
  - Finding the appropriate feature is hard.
  - Doesn't recommend items outside the user profile.

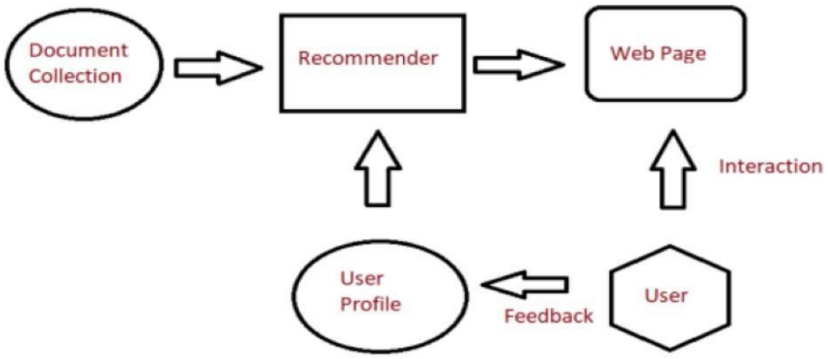

Fig: Recommender System

Pre Lab Exercise:

1. **When can you use content-based recommendation system approach?**
   Content-based recommendation is used when we want to suggest items similar to what the user liked before. It is useful when user preference data is available, like past searches, clicks, or ratings.

   Example: Recommending movies similar to previously watched movies.

2. **Write advantages of content-based filtering approach**
   - It recommends items based on user interest
   - Does not need data from other users
   - Works well for new or unique items
   - Easy to understand and implement
3. **Write disadvantages of content-based filtering approach**
   - Limited recommendations (only similar items)
   - Cannot suggest new or different items
   - Needs detailed item information
   - Overfitting to user's past interest

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Program (Code):

To be attached with

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
df=pd.read_csv("/content/movies.csv")

#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']

#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)

#droping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()
def recommendation_movie(title,topn=5):
  if title not in indices:
    return "Movie is not found!"
  idx=indices[title]
  sim_scores=list(enumerate(cosine_sim[idx]))
  sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
  sim_scores=sim_scores[1:topn+1]
  movie_indices=[i[0] for i in sim_scores]
return df["original_title"].iloc[movie_indices]
recommendation_movie("Inception")
```

Results:

To be attached with

	original_title
2897	Cypher
4401	The Helix... Loaded
134	Mission: Impossible - Rogue Nation
1786	Flatliners
2389	Renaissance

dtype: object

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Observation:

The system recommends movies based on their content such as genres and overview. TF-IDF converts text data into numerical form, and cosine similarity finds how similar the movies are.

When a movie like "Inception" is given as input, the system suggests movies with similar content. It was observed that recommendations are relevant but limited to similar types of movies only.

Post Lab Exercise:

1. **What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.**

   The system can be improved by adding more features like cast, director, and ratings to get better recommendations. Using advanced techniques like machine learning or deep learning can also improve accuracy.

   We can also combine content-based filtering with collaborative filtering to give more diverse and personalized recommendations. Improving the dataset quality and adding user feedback can further enhance the system.

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2. **Recommend top-10 "series" using IMDB movie dataset**
   (Link: https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows). Attach the code with output. Also, write your observation for the same.

Code:

```
import pandas as pd
# Load dataset
df = pd.read_csv("/content/movies.csv")
# Check columns
print(df.columns)
# Remove missing values in rating
df = df.dropna(subset=['vote_average'])
# Sort by rating
top10 = df.sort_values(by='vote_average', ascending=False).head(10)
# Show result
print("\nTop 10 Movies:\n")
print(top10[['original_title', 'vote_average']])
```

Output:

```
Index(['index', 'budget', 'genres', 'homepage', 'id', 'keywords',
       'original_language', 'original_title', 'overview', 'popularity',
       'production_companies', 'production_countries', 'release_date',
       'revenue', 'runtime', 'spoken_languages', 'status', 'tagline', 'title',
       'vote_average', 'vote_count', 'cast', 'crew', 'director'],
      dtype='object')

Top 10 Movies:

                 original_title  vote_average
4662            Little Big Top          10.0
3519          Stiff Upper Lips          10.0
4045     Dancer, Texas Pop. 81          10.0
4247     Me You and Five Bucks          10.0
3992                 Sardaarji           9.5
2386            One Man's Hero           9.3
1881  The Shawshank Redemption           8.5
2970        There Goes My Baby           8.5
3337             The Godfather           8.4
2796     The Prisoner of Zenda           8.4
```